

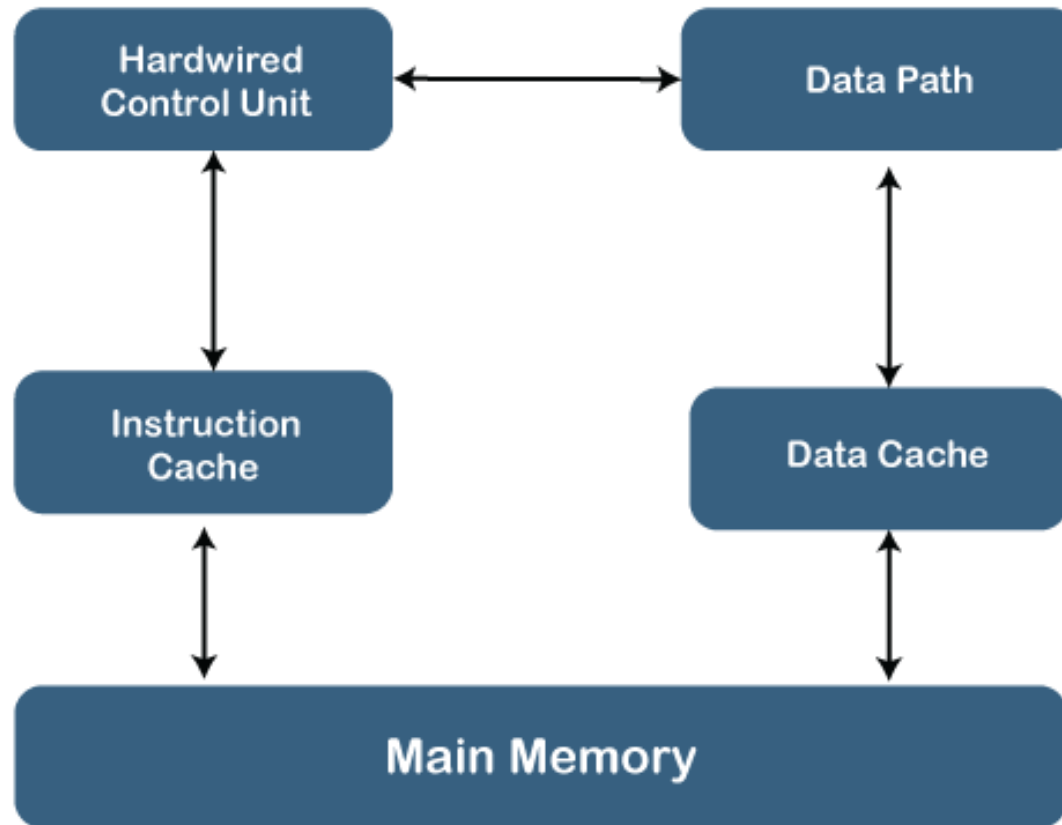
RISC and CISC Architecture

Module 1

RISC

- **RISC :- Reduced Instruction Set Computer Processor**
- A microprocessor architecture with a simple collection and highly customized set of instructions.
- It is built to minimize the instruction execution time by optimizing and limiting the number of instructions.
- It means each instruction cycle requires only one clock cycle, and each cycle contains three parameters: fetch, decode and execute.
- RISC chips require several transistors, making it cheaper to design and reduce the execution time for instruction.

RISC Architecture



RISC Architecture

Features of RISC Processor

- **One cycle execution time:** For executing each instruction in a computer, the RISC processors require one CPI (Clock per cycle).
- **Pipelining technique:** The pipelining technique is used in the RISC processors to execute multiple parts or stages of instructions to perform more efficiently.
- **A large number of registers:** RISC processors are optimized with multiple registers that can be used to store instruction and quickly respond to the computer and minimize interaction with computer memory.
- It supports a simple addressing mode and fixed length of instruction for executing the pipeline.
- It uses LOAD and STORE instruction to access the memory location.
- Simple and limited instruction reduces the execution time of a process in a RISC.

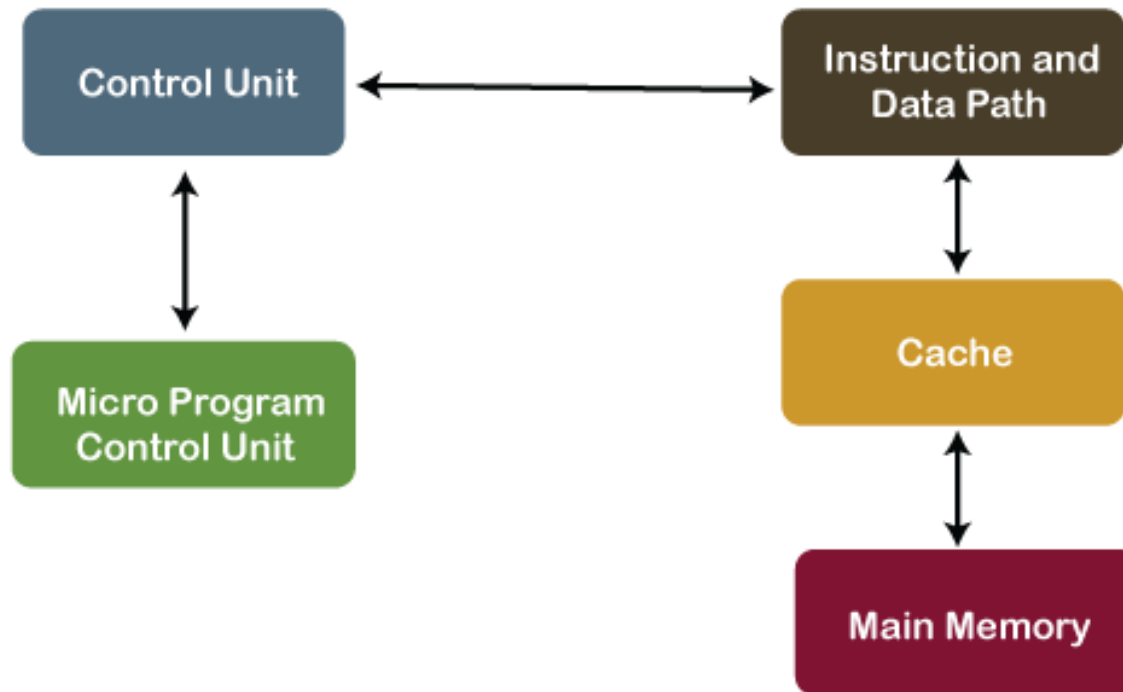
CISC

- **CISC:-Complex Instruction Set Computer**
- It has a large collection of complex instructions that range from simple to very complex and specialized in the assembly language level, which takes a long time to execute the instructions.
- CISC approaches reducing the number of instruction on each program and ignoring the number of cycles per instruction.
- CISC chips are relatively slower as compared to RISC chips but use little instruction than RISC.

Features of CISC Processor

- The length of the code is short, so it requires very little RAM.
- CISC or complex instructions may take longer than a single clock cycle to execute the code.
- Less instruction is needed to write an application.
- It provides easier programming in assembly language.
- Support for complex data structure and easy compilation of high-level languages.
- It is composed of fewer registers and more addressing modes, typically 5 to 20.
- Instructions can be larger than a single word.
- It emphasizes the building of instruction on hardware because it is faster to create than the software.

CISC Architecture



CISC Architecture

Difference between the RISC and CISC Processors

RISC	CISC
It is a Reduced Instruction Set Computer.	It is a Complex Instruction Set Computer.
It emphasizes on software to optimize the instruction set.	It emphasizes on hardware to optimize the instruction set.
It is a hard wired unit of programming in the RISC Processor.	Microprogramming unit in CISC Processor.
It requires multiple register sets to store the instruction.	It requires a single register set to store the instruction.
RISC has simple decoding of instruction.	CISC has complex decoding of instruction.
Uses of the pipeline are simple in RISC.	Uses of the pipeline are difficult in CISC.
It uses a limited number of instruction that requires less time to execute the instructions.	It uses a large number of instruction that requires more time to execute the instructions.
It uses LOAD and STORE that are independent instructions in the register-to-register a program's interaction.	It uses LOAD and STORE instruction in the memory-to-memory interaction of a program.
RISC has more transistors on memory registers.	CISC has transistors to store complex instructions.
The execution time of RISC is very short.	The execution time of CISC is longer.
RISC architecture can be used with high-end applications like telecommunication, image processing, video processing, etc.	CISC architecture can be used with low-end applications like home automation, security system, etc.
It has fixed format instruction.	It has variable format instruction.
The program written for RISC architecture needs to take more space in memory.	Program written for CISC architecture tends to take less space in memory.
Example of RISC: ARM, PA-RISC, Power Architecture, Alpha, AVR, ARC and the SPARC.	Examples of CISC: VAX, Motorola 68000 family, System/360, AMD and the Intel x86 CPUs.